HJM model density of the shifted lognormal case with stochastic volatility

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1 Model formulation

Starting from the multi-factor HJM term structure model for the dynamics of the forward rates, driven by N correlated Brownian motions, we have

$$df(t,T) = \left(\sum_{i,j} \sigma_i(t,T)\rho_{ij}(t) \int_t^T \sigma_j(t,u)\,du\right) dt + \sum_i \sigma_i(t,T)\,dW_i(t). \tag{1}$$

Under the separability condition

$$\sigma_i(t,u) = \sigma_i(t,X)\,\theta_i(u) \tag{2}$$

we obtain

$$\begin{aligned} df(t,T) &= \sum_{i,j} \theta_i(T)\sigma_i(t,X)\rho_{ij}(t)\sigma_j(t,X)\left(\int_t^T \theta_j(u)\,du\right) dt + \sum_i \theta_i(T)\sigma_i(t,X)\,dW_i(t) \\ &= \sum_{i,j} \theta_i(T)\,d\Psi_{i,j}(t,X)\,\Theta_j(T) + \sum_i \theta_i(T)\,dX_i(t), \end{aligned}$$

where

$$\Theta_j(T) = \int_{t_0}^T \theta_j(u)\,du, \tag{3}$$

$$d\Psi_{i,j}(t,X) = \sigma_i(t,X)\rho_{ij}(t)\sigma_j(t,X)\,dt, \tag{4}$$

$$dX_i = \sigma_i(t,X)\left(dW_i(t) - \sum_j \rho_{ij}(t)\sigma_j(t,X)\Theta_j(t)\,dt\right), \tag{5}$$

with $X_i(t_0) = 0$ and $\Psi_{i,j}(t_0) = 0$.

Integrating then yields

$$\begin{aligned} f(t,T) &= f(t_0,T) + \sum_i \theta_i(T)\left(X_i(t) + \sum_j \Psi_{i,j}(t,X)\Theta_j(T)\right), \\ r_t &= f(t_0,t) + \sum_i \theta_i(t)\left(X_i(t) + \sum_j \Psi_{i,j}(t,X)\Theta_j(t)\right) \\ &= f(t_0,t) + \overline{r}_t, \\ P(t,T) &= \exp\left(-\int_t^T f(t_0,u)\,du\right), \\ B(t,T) &= P(t,T)Z_t \exp\left(-\sum_i \Theta_i(T)\left(X_i(t) + \frac{1}{2}\sum_j \Psi_{i,j}(t,X)\Theta_j(T)\right)\right), \end{aligned}$$

where

$$Z_t \;=\; \exp\left(\sum_i \Theta_i(t)\left(X_i(t) + \frac{1}{2}\sum_j \Psi_{i,j}(t,X)\Theta_j(t)\right)\right).$$

Introduce the Radon–Nikodym process M_t by

$$\frac{dM_t}{M_t} \;=\; \sum_i \Theta_i(t)\sigma_i(t)\,dW_i(t). \tag{6}$$

Theorem 1.1. *The Radon–Nikodym derivative M_t satisfies*

$$\frac{M_t}{M_s} \;=\; \exp\left(-\int_s^t \overline{r}_u\,du\right)\frac{Z_t}{Z_s}.$$

Under this measure the dynamics become

$$\begin{aligned}
\Theta_j(T) &= \int_{t_0}^T \theta_j(u)\,du,\\
d\Psi_{i,j}(t,X) &= \sigma_i(t,X)\rho_{ij}(t)\sigma_j(t,X)\,dt,\\
dX_i &= \sigma_i(t,X)\,dW_i^M(t),
\end{aligned}$$

and

$$E_t^Q\left[\exp\left(-\int_t^T r_u\,du\right)\Phi(X_T,\Psi_T)\right] \;=\; P(t,T)Z_t\,E_t^M\left[\frac{\Phi(X_T,\Psi_T)}{Z_T}\right].$$

Proof. Itô's formula gives

$$\Theta_i(t)X_i(t) - \Theta_i(s)X_i(s) \;=\; \int_s^t \theta_i(u)X_i(u)\,du + \int_s^t \Theta_i(u)\,dX_i(u).$$

Hence

$$\begin{aligned}
\int_s^t \overline{r}_u\,du &= \sum_i\left(\Theta_i(t)X_i(t) - \Theta_i(s)X_i(s) - \int_s^t \Theta_i(u)\sigma_i(u)\,dW_i(u)\right)\\
&\quad + \sum_{i,j}\int_s^t \theta_i(u)\Psi_{i,j}(u)\Theta_j(u)\,du + \sum_{i,j}\int_s^t \Theta_i(u)\sigma_i(u)\rho_{ij}(u)\sigma_j(u)\Theta_j(u)\,du.
\end{aligned}$$

Integration by parts yields

$$\begin{aligned}
\sum_{i,j}\int_s^t \theta_i(u)\Psi_{i,j}(u)\Theta_j(u)\,du &= \frac{1}{2}\sum_{i,j}\Big(\Theta_i(t)\Psi_{i,j}(t,X)\Theta_j(t) - \Theta_i(s)\Psi_{i,j}(s,X)\Theta_j(s)\Big)\\
&\quad -\frac{1}{2}\sum_{i,j}\int_s^t \Theta_i(u)\sigma_i(u)\rho_{ij}(u)\sigma_j(u)\Theta_j(u)\,du.
\end{aligned}$$

Therefore

$$\begin{aligned}
\int_s^t \overline{r}_u\,du &= \sum_i \Theta_i(t)\left(X_i(t) + \frac{1}{2}\sum_j \Psi_{i,j}(t,X)\Theta_j(t)\right)\\
&\quad - \sum_i \Theta_i(s)\left(X_i(s) + \frac{1}{2}\sum_j \Psi_{i,j}(s,X)\Theta_j(s)\right)\\
&\quad - \sum_i \int_s^t \Theta_i(u)\sigma_i(u)\,dW_i(u) + \frac{1}{2}\sum_{i,j}\int_s^t \Theta_i(u)\sigma_i(u)\rho_{ij}(u)\sigma_j(u)\Theta_j(u)\,du.
\end{aligned}$$

The first two lines are $\log Z_t - \log Z_s$. The remaining stochastic exponential is exactly M_t/M_s, which proves

$$\frac{M_t}{M_s} = \exp\left(-\int_s^t \overline{r}_u\, du\right)\frac{Z_t}{Z_s}.$$

Under M, the finite-variation drift in dX_i is removed by Girsanov, and the pricing identity follows from $dQ|_{\mathcal{F}_T}/dM|_{\mathcal{F}_T} = M_T^{-1}$ together with $P(t,T) = \exp(-\int_t^T f(t_0,u)\,du)$. □

2 Shifted lognormal case with lognormal stochastic volatility

In this section we consider the one-factor case with volatility

$$\sigma(t,X) = \sigma(t)\,\eta_t\left(\beta(X+\gamma\Psi(t,X))+1\right), \tag{7}$$

where η_t is a positive process, independent of W. (If $\mathrm{d}\langle\eta,W\rangle \neq 0$, the time-changed Brownian motion is no longer independent of the clock and the Matsumoto–Yor formulae below do not apply.)

Set

$$\begin{aligned} S_t &= \beta(X+\gamma\Psi(t,X))+1, \\ A_t &= \beta^2\Psi(t,X). \end{aligned}$$

The initial conditions $X(t_0) = \Psi(t_0) = 0$ give $S_{t_0} = 1$ and $A_{t_0} = 0$. Using Theorem 1.1 under the M-measure we have

$$\frac{dS_t}{S_t} = \beta\sigma(t)\,\eta_t\,dW_t^M + \beta\gamma\,\sigma^2(t)\,\eta_t^2\,S_t\,dt, \tag{8}$$

$$dA_t = \beta^2\sigma^2(t)\,\eta_t^2\,S_t^2\,dt. \tag{9}$$

Define the stochastic time change

$$d\tau_t = \beta^2\sigma^2(t)\,\eta_t^2\,dt,$$

so that

$$\tau_t = \beta^2\int_{t_0}^t \sigma^2(u)\,\eta_u^2\,du. \tag{10}$$

Then, under M and in the time-changed clock,

$$\frac{dS_\tau}{S_\tau} = dW_\tau^M + \frac{\gamma}{\beta}\,S_\tau\,d\tau, \tag{11}$$

$$dA_\tau = S_\tau^2\,d\tau. \tag{12}$$

Girsanov with kernel $(\gamma/\beta)S$ removes the finite-variation drift. The resulting density is

$$L_\tau := \frac{dQ^D}{dQ^M} = \exp\left(-\frac{\gamma}{\beta}(S_\tau - 1) + \frac{\gamma^2}{2\beta^2}A_\tau\right),$$

hence

$$\frac{dQ^M}{dQ^D} = L_\tau^{-1} = \exp\left(\frac{\gamma}{\beta}(S_\tau - 1) - \frac{\gamma^2}{2\beta^2} A_\tau\right).$$

Under Q^D the dynamics become

$$\begin{aligned} S_\tau &= \exp\left(W_\tau^D - \frac{\tau}{2}\right), \\ A_\tau &= \int_0^\tau S_u^2\, du, \\ U_\tau &= \frac{S_\tau}{A_\tau}. \end{aligned}$$

In particular $S_\tau = \exp(B_\tau^{(-1/2)})$ and $A_\tau = A_\tau^{(-1/2)}$ in the notation of Matsumoto and Yor [1]. Pricing identities under M are rewritten under D by

$$E^M\left[\frac{g}{Z}\right] = E^D\left[L^{-1}\frac{g}{Z}\right].$$

The true one-factor HJM bond is, with $X = (S-1)/\beta - \gamma A/\beta^2$ and $\Psi = A/\beta^2$,

$$\frac{B(t,T)}{Z_t} = P(t,T)\exp\left(-\frac{\Theta(T)}{\beta}(S_\tau - 1) + \frac{\gamma\Theta(T)}{\beta^2}A_\tau - \frac{\Theta(T)^2}{2\beta^2}A_\tau\right).$$

Multiplying by the Radon–Nikodym factor L^{-1} produces the D-measure kernel

$$L_\tau^{-1}\frac{B(t,T)}{Z_t} = P(t,T)\exp\left(-\left(\frac{\Theta(T)-\gamma}{\beta}\right)(S_\tau - 1) - \frac{A_\tau}{2}\left(\frac{\Theta(T)-\gamma}{\beta}\right)^2\right).$$

(The previous display is *not* the bond B/Z; it is the integrand for an expectation under Q^D.)

Write $\alpha(T) = (\Theta(T)-\gamma)/\beta$. The joint law of (S_τ, U_τ) under Q^D, conditionally on the clock τ (i.e. on the path of η), follows from [1, Thm. 4.1] after the change of variables $U = S/A$. The conditional law of S given $U = u$ is inverse Gaussian with mean 1 and shape u,

$$p(S_\tau = x \mid U_\tau = u, \tau) = \sqrt{\frac{u}{2\pi x^3}}\exp\left(u - \frac{u}{2}\left(x + \frac{1}{x}\right)\right),$$

and the marginal of U is

$$p(U_\tau = u \mid \tau) = \sqrt{2\pi}\, \exp\left(-u - \frac{\tau}{8}\right)\frac{H(u,\tau)}{u^{3/2}},$$

where H is the Hartman–Watson function θ of [1]. The conditional density in S does not depend on τ.

The product of this density with the D-kernel, divided by $P(t,T)$, is the density of an inverse Gaussian law with mean $1/\mu$ and shape U_τ, where $\mu = 1 + \alpha(T)/U_\tau = 1 + (\Theta(T) - \gamma)/(\beta U_\tau)$:

$$\begin{aligned}
p(x \mid U_\tau, \tau)\, L^{-1} \frac{B(t,T)}{P(t,T) Z_t}\bigg|_{S=x} &= \sqrt{\frac{U_\tau}{2\pi x^3}} \exp\left(\alpha(T) + U_\tau - \frac{1}{2}\left(\frac{x}{U_\tau}\Big(U_\tau + \alpha(T)\Big)^2 + \frac{U_\tau}{x}\right)\right) \\
&= \sqrt{\frac{U_\tau}{2\pi x^3}} \exp\left(-\frac{U_\tau}{2x}\left(x\mu - 1\right)^2\right) \\
&= \sqrt{\frac{U_\tau}{2\pi x^3}} \exp\left(-\frac{1}{2}\left(\sqrt{U_\tau x}\,\mu - \sqrt{\frac{U_\tau}{x}}\right)^2\right) \\
&= \sqrt{\frac{U_\tau}{2\pi x^3}} \exp\left(2\mu U_\tau - \frac{1}{2}\left(\sqrt{U_\tau x}\,\mu + \sqrt{\frac{U_\tau}{x}}\right)^2\right).
\end{aligned}$$

(The four right-hand sides are identical; they are a density times a kernel, not the bond itself.)

Define

$$\begin{aligned}
\xi^+(k) &= \sqrt{U_\tau k}\,\mu - \sqrt{\frac{U_\tau}{k}}, \\
\xi^-(k) &= -\sqrt{U_\tau k}\,\mu - \sqrt{\frac{U_\tau}{k}}.
\end{aligned}$$

At frozen U_τ one has $d\xi^+ + d\xi^- = \sqrt{U_\tau/k^3}\,dk$, but the two branches carry different Jacobians; the Gaussian split uses each branch separately. Integrating the product against $\mathbf{1}_{\{S \le k\}}$ yields the inverse-Gaussian cdf (mean $1/\mu$, shape U_τ)

$$E^D\left[L^{-1}\frac{B(t,T)}{Z_t}\mathbf{1}_{\{S_\tau \le k\}}\,\bigg|\, U_\tau, \tau\right] = P(t,T)\left(N\big(\xi^+(k)\big) + \exp\left(2\left(U_\tau + \frac{\Theta(T)-\gamma}{\beta}\right)\right) N\big(\xi^-(k)\big)\right).$$

A payer swaption with expiry T_e, last payment T_n and strike K has time-T_e value $(1 - B(T_e,T_n) - K\sum_{i=e+1}^n \delta_i B(T_e,T_i))^+$, using $B(T_e,T_e) = 1$. Theorem 1.1 gives

$$\begin{aligned}
\text{swap}(t) &= P(t,T_e) Z_t\, E^M\left[\frac{\Big(1 - B(T_e,T_n) - K\sum_{i=e+1}^n \delta_i B(T_e,T_i)\Big)^+}{Z_{T_e}}\right] \\
&= P(t,T_e) Z_t\, E^D\left[L^{-1}\Big(G_e - G_n - K\sum_{i=e+1}^n \delta_i G_i\Big)^+\right],
\end{aligned}$$

where $G_i := B(T_e,T_i)/Z_{T_e}$. Given U, every G_i is strictly monotone in S, so the forward swap value has a unique root S^* and the positive part is the same linear combination restricted to $\{S > S^*\}$ (or $\{S < S^*\}$, according to the sign of β). Each term $E^D[L^{-1}G_i\mathbf{1}_{\{S \gtrless S^*\}} \mid U, \tau]$ is the inverse-Gaussian formula above with tenor-specific $\alpha(T_i) = (\Theta(T_i) - \gamma)/\beta$. The remaining average is over the Hartman–Watson law of U given τ and over the law of the clock $\tau = \beta^2\int_{t_0}^{T_e}\sigma^2(u)\eta_u^2\,du$. If η^2 is CIR, the mixing law of τ is that of Dufresne [3].

References

[1] H. Matsumoto and M. Yor, Exponential functionals of Brownian motion, I: Probability laws at fixed time, *Probability Surveys* **2** (2005), 312–347.

[2] L. B. G. Andersen and V. V. Piterbarg, *Interest Rate Modeling. Volume I: Foundations and Vanilla Models*, Atlantic Financial Press, 2010.

[3] D. Dufresne, The integrated square-root process, working paper, University of Melbourne, 2001.